Kirya Caine

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EDUCATION

Bachelor of Science in Neuroscience, *University of Rochester,* *Rochester NY.* December 2025
***Magna Cum Laude*, Highest Honors in Neuroscience, Dean's List all semesters. GPA 3.95/4.00.**
Relevant coursework:

Neuroscience: Neurochemical Basis of Behavior, Pharmacology, Bio of Mental Disorders, Neurobio of sleep, Developmental Psychopathology, Biochemistry, Organic Chemistry I & II
Evolution, Ethology: Neuroethology, Evolution, Human Evolutionary Genetics, Animal Minds, Darwin and Darwinism
Skills for Research: Grant Writing, Programming and Statistics, Data Science, Lit. Review

FELLOWSHIPS, GRANTS, AND HONORS

- UCSF NIH-PREP Fellowship (2026 – $56000)
- Stanford Undergraduate Research Conference (SRC 2026) Outstanding Poster Award in Life Sciences
- Stanford Undergraduate Research Conference Travel Award (2026 – $400)
- Barry Goldwater Scholarship Nominee (2025)
- Recny Fellowship in Biological Sciences (Summer 2025 – $5000)
- University of Rochester Discover Grant (Summer 2025 – $5000, declined for Recny Fellowship)
- Irene and Eric Simon Brain Research Foundation Research Fellowship (Summer 2024 – $2500)
- Whipple Science and Research Scholarship Recipient (2022–2025 – $52000)

RESEARCH EXPERIENCE

NIH-PREP Postbaccalaureate Research Fellow and Junior Specialist, Advisor: Dr. Corey Harwell
UCSF, Department of Neurology July 2026 – Present

- **Formulated, proposed, and currently executing a project** leveraging single-nucleus RNA-seq data to map the differentiation trajectories and identify transcriptional disruptions of lateral septum (LS) oligodendrocytes following prenatal opioid exposure and neonatal opioid withdrawal syndrome (NOWS). Candidate differentially expressed genes (DEGs) will be validated using RNAscope in situ hybridization to gain novel insight into the mechanisms of LS glial maturation and ramifications of developmental perturbation.
    - Supporting multiple ongoing projects in the lab by performing perfusions, precision microtomy, confocal microscopy, and maintaining transgenic mouse colony
    - Submitting 2027 GRFP based on work in this lab (see Research Proposals and Scientific Writing)

Student Researcher, Advisor: Dr. Manoela Fogaça
University of Rochester, Department of Pharmacology & Physiology August 2024 – July 2026

- **Independently designed, proposed, and completed a 2-year project** using behavioral neuropharmacology and histology to explore the impact of α5-GABAA receptor positive allosteric modulation on cortico-hippocampal synaptic protein expression, plasticity markers, and adaptive behaviors in stressed and naive mice.
    - Authored first-author manuscript (in preparation) detailing sex-divergent and region-specific neurobiological and behavioral outcomes
    - Supported by the Recny Fellowship. Gave award-winning poster presentation of cumulative work at SRC 2026 and institutional UR symposia. Upcoming poster at ABRCMS 2026.
    - Presented several lab-wide journal clubs focusing on the relationship between cortico-hippocampal excitation-inhibition balance and the mesolimbic dopamine system.
    - Authored section of a Review (in prep) on excitation/inhibition balance in depression pathology and emerging treatments

Undergraduate Research Assistant, Advisor: Dr. Jean Bidlack
University of Rochester, Department of Pharmacology & Physiology August 2023 – August 2024

- **Main Project:** Executed in vitro radioligand competition assays to screen and evaluate affinity of novel ligands for κ and μ opioid receptors, screening potential novel pharmacotherapies to combat cocaine abuse and relapse through the κOR
- **Summer Project:** Received the Irene and Eric Simon Foundation's 2024 Summer Research Fellowship in Neuroscience to investigate Fibroblast Growth Factor 21 (FGF21) as a novel avenue for understanding cocaine reward using conditioned place preference in FGF21-overexpressing transgenic mice. Prepared a final report and virtual oral presentation for the IES Foundation

Young Research Scholar, Advisor: Dr. Devesh Tiwari
Northeastern University, Department of Electrical & Computer Engineering Summer 2021

- **Main Project:** Conducted a literature review and gave department-wide presentation exploring the applications of quantum computing to protein-folding and pharmacological discovery research

PUBLICATIONS

Caine K, Daher F, Ingebretsen E, Fogaça M. *Positive Allosteric Modulation of α5-subunit containing GABAA Receptors Exerts Sex-Specific Effects on Sociality, Cognition, and Despair, but not Anhedonia.* Manuscript in preparation

Daher F, Ingebretsen E, Fukushima TC, **Caine K**, Bazhenov N, MacLean DM, Singh B, Luscher B, Fogaça M. *Rebalancing excitation and inhibition: The Glutamatergic–GABAergic framework for rapid antidepressant mechanisms*. Manuscript in preparation (personal section completed); targeted for submission late 2026 (Review)

Bischoff MA, Wang X, Song E, Wilson G, Brody M, Iyer S, **Caine K**, Clark M. *Martha Chase at the University of Rochester: The Woman in STEM Who Was Forgotten. The University of Rochester Journal of Undergraduate Research*. Spring 2024. https://doi.org/10.47761/HEMA1238

POSTER PRESENTATIONS

Caine K, Daher F, Fogaça M. *Rapid and Selective Antidepressants*: *α5-GABAAR Positive Allosteric Modulation Promotes Adaptive Behavior and Cortical VGAT upregulation in mice*

- Stanford Undergraduate Research Conference 2026; Award for Outstanding Poster in Life Sciences
- Upcoming Poster at ABRCMS 2026
- University of Rochester Undergraduate Program in Biology and Medicine 2025 Poster Session

Caine K, Daher F, Fogaça M. *α5-GABAAR Positive Allosteric Modulation Induces Rapid and Sustained Antidepressant-Like Effects*. University of Rochester 2025 Research Exposition

TEACHING

BIOL 205 – Evolution, Teaching Assistant (Spring 2024), Lead Teaching Assistant (Spring 2025)
Responsibilities: Assisting ~50 students per semester, planning and leading 2 recitations per week covering homework and journal articles, grading exams and weekly homework, leading review sessions, holding office hours.

- In 2025, advised new faculty, Dr. Karl Glastad, in curriculum design, content selection, and exam writing for his first time teaching the course.
- I designed mini-lectures to demystify difficult concepts and worked one-on-one with struggling students to bring their weekly assignment averages from ~50% to full marks.

NSCI 244 – Neuroethology, Teaching Assistant, Spring 2025.
Responsibilities: Assisting ~40 students, hosting journal clubs, grading weekly homework, leading review sessions before exams, holding office hours

NSCI 201P – Basic Neurobiology Lab, Teaching Assistant, Fall 2025.
Responsibilities: Assisting ~20 students, providing in-lab hands-on support and instruction for

anatomical dissections, electrophysiology, neuropharmacology, and behavioral conditioning assays. Grading post-lab assignments

LEADERSHIP AND SERVICE

Undergraduate keynote speaker at the *For Ever Better Campaign Launch*, Meliora Weekend 2025. Spoke to 500+ Board of Trustee Members and honored donors involved in the $1.75 billion Campaign. Sole undergraduate selected. Presented my research and cultural leadership, evaluating how these synergize to improve the academic community. Advocated for merit scholarships and research grants.

Filipino American Students' Association (FASA), President (2025 – 2026), Secretary (2024 – 2025). Cultural organization working to preserve and share Filipino culture with the University of Rochester community through weekly general member meetings, monthly cooking sessions, and large-scale events.

- Doubled event attendance by improving campus-wide marketing, introducing a "Cook with FASA" program, and reviving a Tinikling Performance Group (traditional Filipino dance)
- Formalized early enrollment in "Ate/Kuya" mentorship program to help new students find community and pre-professional support, organized annual events to expand and strengthen alumni network
- Implemented a new semester-long scheduling system which cut weekly e-board meeting duration by >50% of previous average and reduced the number of late event registrations from ~4/semester to 0.
- Earned Awards for club achievement and engagement, totaling $700

Organizing Committee Member, Outreach, Scientific Innovation Through Different Perspectives (SITDP) Conference (2026 – 2027). Annual national conference where postbaccalaureate scholars from PREP and PROPEL present their research and access professional development programming. Serving on the outreach committee, recruiting and coordinating speakers, panelists, and poster judges.

Research Panelist, UCSF PREP (2026 – Present). Panelist for program events, answering undergraduate questions about finding and applying to research positions, choosing between postbaccalaureate programs and direct graduate applications, and preparing competitive applications.

Senior Biology Peer Advisor (2025 – 2026). Paid position for 5 seniors majoring in biology who have demonstrated outstanding achievement. Held meetings with prospective and current students providing insight into searching for/applying to research positions and grants, course selection, and resume building.

Neuroscience Representative/Panelist, University of Rochester Undergraduate Research Fair and Society for Undergraduate Biology Students (2023 – 2026). Panelist for annual university-wide events to answer attendee questions about selecting, applying to, and doing work in a Neuroscience lab.

Student Alumni Ambassador (2025 – 2026). Selected to represent the University for my community leadership and academic record. Served as a liaison to alumni, major donors, and prospective students by hosting on-campus, local, and regional events and coordinating networking efforts.

Indigo Fire Pottery Studio, Summer Program Coordinator and Studio Technician (2018 – 2024). Head of Summer Ceramics Education Program. Designed lesson plans, program schedule, and taught 150+ students ceramics. Trained staff and managed studio operations, including executing kiln firings of 2000° F and optimizing and executing protocols for large-scale glaze chemistry.

RESEARCH PROPOSALS AND SCIENTIFIC WRITING

Wiring the Social Brain: Experience-Dependent Oligodendrocyte Maturation and Vasopressin Circuit Assembly in the Lateral Septum. Proposed and Designed NSF Graduate Research Fellowship Proposal leveraging single-nucleus transcriptomics to investigate the effects of early life social

deprivation and enrichment on oligodendrocyte maturation, myelination, and vasopressin circuit assembly during development in the Lateral Septum, a conserved node of the Social Behavior Network.

Using Epigenetics to Explore Novel Treatments for Alcohol Abuse: Histone Modification in the Hippocampus and Amygdala. Synthesized literature on how alcohol-induced alterations in limbic histone acetylation and microRNA regulation drive withdrawal and relapse behaviors. Proposed novel therapeutic strategies targeting histone modification pathways and epigenetic feedback loops in AUD.

***Evolution and Mechanisms of Strategic Sibling Aggression in Masked and Blue-Footed Booby Birds*.** Designed a mock grant proposal investigating the drivers of facultative versus obligate siblicide across closely related seabird species. Integrated evolutionary, ethological and neuroendocrine frameworks to propose testable hypotheses and design experiments.

Astrocytes, Aquaporin-4, and Alzheimer's: The Emergence of a New Therapeutic Target in AD. Evaluated molecular, behavioral, and clinical evidence linking AQP4 dysfunction to amyloid and tau accumulation through glymphatic impairment, proposing astrocytes as a therapeutic target for AD.

Sleeping with the Fishes: Understanding Sleep Architecture Through Comparative Biology in Marine Mammals. Integrated ethological, electrophysiological, and genetic studies to address contemporary debates regarding phylogenetic, ecological, and functional influences on sleep structure.

LABORATORY SKILLS

- **Behavioral neuroscience:** Perfusion and dissection of mice, husbandry, Forced Swim Test, Sucrose Splash Test, Open Field Test, Tail Suspension Test, Female Urine Sniffing Test, Novel Object Recognition Test, Social Interaction Test, Elevated Plus Maze, Conditioned Place Preference, Chronic Unpredictable Stress
- **Molecular biology and histology:** Brain sectioning using cryostat and vibratome, immunohistochemistry, RNAscope, Fluorescent and Confocal microscopy, Genotyping/PCR, Radioligand binding assays, BRET (bioluminescence Resonance Energy Transfer)
- **Data management, analysis, and visualization:** Bioinformatics in R (including snRNAseq analysis using Seurat: dimensionality reduction, trajectory inference), MATLAB, Prism, Ethovision, Sigmaplot